Stochastic Eigenvalue Estimation via the Rodeo Algorithm: Application to Ising and Discrete Dirac Hamiltonians

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Abstract

We implement and benchmark the *rodeo algorithm* [1] for quantum eigenvalue estimation using the Qiskit framework. The algorithm is applied to the one-dimensional transverse-field Ising model ($L = 4$ spins) under three noise conditions: (i) ideal simulation, (ii) custom depolarising noise, and (iii) a realistic IBM Manila hardware profile. We also present a generalised implementation targeting the one-dimensional naive discrete Dirac Hamiltonian, demonstrating that the rodeo filter can resolve the lattice fermion spectrum. All source code and simulation scripts are openly available on GitHub at `https://github.com/ArthurFCSMello/Rodeo_project`.

1 Introduction

Extracting eigenvalues and eigenstates of many-body Hamiltonians is a central problem in quantum physics and chemistry. Quantum Phase Estimation (QPE) is the textbook approach, but its circuit depth scales as $O(1/\varepsilon)$, making it prohibitive on near-term devices.

Choi *et al.* [1] introduced the **rodeo algorithm**, a stochastic filter that converges to a target eigenvector with exponential accuracy in the number of ancilla measurements. For eigenvalue determination with error ε the cost is $O\big[(\log\varepsilon)^2/(p\,\varepsilon)\big]$, where p is the squared overlap of the initial state with the target eigenvector.

In this work we:

1. reproduce the rodeo spectrum of the transverse-field Ising model,
2. quantify the effect of depolarising and hardware-level noise on the spectral signal, and
3. generalise the circuit to the one-dimensional discrete Dirac Hamiltonian relevant for lattice fermion studies.

It is crucial to emphasise that despite the rich and varied physical interpretations of these models—such as the emergence of antimatter in the Dirac equation or bound states in ferromagnetic islands for the spin chains—the computational core of the algorithm remains fundamentally unchanged. The essence of the approach is the quantum simulation of quantum systems by framing them as generalised eigenvalue and eigenvector extraction problems.

2 Theoretical Framework

2.1 The Rodeo Filter

Consider a Hamiltonian H with eigenstates $|E_n\rangle$ and eigenvalues E_n. An ancilla qubit is prepared in $|+\rangle$ and couples to the system via controlled time evolution $CU(t) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes e^{-iHt}$. After applying a phase $e^{iE_{\text{target}}t}$ on the ancilla and a final Hadamard, measuring the ancilla in $|1\rangle$ occurs with probability

$$P_k(E_n) = \cos^2\Big[\frac{(E_n - E_{\text{target}})\,t_k}{2}\Big]. \tag{1}$$

After N independent cycles with random times $\{t_k\}$ the joint success probability is

$$P(E_n) = \prod_{k=1}^{N} \cos^2\Big[\frac{(E_n - E_{\text{target}})\,t_k}{2}\Big]. \tag{2}$$

This function peaks sharply at $E_{\text{target}} = E_n$ and is exponentially suppressed elsewhere, acting as a stochastic spectral filter. The times t_k are drawn from a Gaussian distribution with standard deviation t_{rms}.

2.2 Transverse-Field Ising Model

The Hamiltonian on L sites with open boundary conditions is

$$H_{\text{Ising}} = -J \sum_{i=1}^{L-1} Z_i Z_{i+1} - h \sum_{i=1}^{L} X_i \,, \tag{3}$$

with coupling $J = 1.0$ and transverse field $h = 1.5$. We use first-order Trotterisation to decompose $e^{-iH_{\text{Ising}}t} \approx \left(e^{iht\sum X/n}\, e^{iJt\sum ZZ/n}\right)^n$, implementing each factor with controlled-R_x and CNOT–controlled-R_z–CNOT gadgets.

2.3 1D Discrete Dirac Hamiltonian

The naive discrete Dirac Hamiltonian on a 1D lattice of spacing a reads

$$H_D = \frac{1}{2ai} \sum_n \sigma_z \big(|n{+}1\rangle\langle n| - |n{-}1\rangle\langle n|\big) + m\,\sigma_x \,, \tag{4}$$

where $\sigma_{x,z}$ act on the two-component spinor (chirality) index. Position $n = 0, \ldots, L{-}1$ is encoded in $\log_2 L$ qubits; the spinor in one additional qubit. For $L = 4$ sites with periodic boundaries, the momentum spectrum takes discrete values $k_j = 2\pi j/(La)$, yielding exactly four eigenvalues: $E(k_j) = \pm\sqrt{(\sin(k_j a)/a)^2 + m^2}$. The hopping operator $|n{+}1\rangle\langle n|$ is realised as a controlled-increment circuit.

3 Implementation

All circuits are built with **Qiskit v2** and executed on the `AerSimulator` backend. The basic structure of the rodeo filter cycle with a single ancilla qubit is depicted in Figure 1.

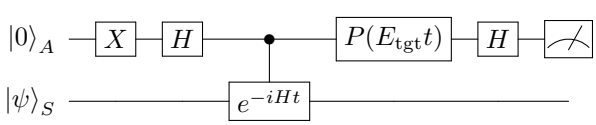

Figure 1: Illustrative quantum circuit for a single rodeo cycle. The ancilla (top) controls the time evolution of the system register (bottom) over a sampled duration t, before accumulating the target energy phase and being measured. The *conditional reset* trick for single-ancilla hardware reuse applies an X and a phase correction if measured in $|1\rangle$.

The time evolution e^{-iHt} is approximated with Trotterisation. The key building block for the Ising model is shown in Listing 1.

To execute the algorithm efficiently on current hardware topologies, we utilise a *single-ancilla* approach with mid-circuit measurements and conditional resets, as detailed in Listing 2.

Circuit Depth Considerations. The rodeo filter has an intrinsic $O(N \cdot n_{\text{Trotter}} \cdot L)$ hardware scaling. For the $L = 4$ Ising model with 20 Trotter steps and 10 cycle iterations, each sequential circuit contains over 1 200 CNOT gates (three CNOTs or controlled-rotations per term slice). This immense depth explains the severe signal degradation when exposed to physical hardware noise profiles. There is an inherent algorithmic trade-off here: increasing n_{Trotter} improves theoretical commutation precision, but the extra sequential depth amplifies physical hardware noise, which can completely flatten the probabilistic signal in practice.

For the Dirac algorithm, we build the kinetic hopping using controlled increments (Listing 3), which involves multi-controlled X (MCX) gates. On restricted hardware topologies, transpiling a single MCX gate requires complex decompositions into dozens of CNOTs, inflating the circuit depth drastically compared to the Ising model.

For the noiseless studies (Figs. 3 and 4), we simulate the ideal filter using 20 independent realisations matching an equivalent of 3 000 shots per point across $E \in [-8, 8]$. For the noisy execution (Fig. 5), we use 10 cycles, 50 energy scan points, and exactly 1 000 shots sampled from the respective physical noise profile. Times are drawn from $\mathcal{N}(0,\, t_{\text{rms}}{=}5)$.

Noise models. Two noise scenarios are investigated:

- **Depolarising:** $p_{1q} = 0.1\%$, $p_{2q} = 1\%$ applied to all single- and two-qubit gates respectively.
- **IBM Manila:** The full noise profile (T_1/T_2 relaxation, readout errors, gate errors) extracted from `FakeManilaV2`, a calibrated snapshot of the 5-qubit IBM Falcon processor.

The Manila simulation uses a *single-ancilla* circuit with mid-circuit measurement and conditional reset to fit within the five-qubit topology.

4 Results

4.1 Noiseless Ising Spectrum

Figure 3 shows the success probability $P(E_{\text{target}})$ for the ideal simulation alongside the evolution of the spectrum reconstruction (Fig. 2) as the number of ancilla cycles N is increased.

Clear peaks emerge at the eigenvalues of H_{Ising}, with the dominant peaks near $E \approx -5.0$, -2.2, and $+1.7$ matching the low-lying eigenvalues of the $L{=}4$ TFIM at $J{=}1$, $h{=}1.5$.

Increasing the scan resolution from 200 to 300 points (Figure 4) improves the peak localisation without affecting the peak heights, confirming convergence of the filter.

4.2 Effect of Noise

Under depolarising noise (Figure 5), the spectral peaks are broadened and suppressed. With 1 000 shots and 50 scan points the signal-to-noise ratio degrades significantly,

```python
# Transverse-field term: -h X_i
for i in range(num_qubits):
    qc.crx(-2 * h * dt, anc, sys[i])
# Ising coupling term: -J Z_i Z_{i+1}
for i in range(num_qubits - 1):
    qc.cx(sys[i], sys[i+1])
    qc.crz(-2 * J * dt, anc, sys[i+1])
    qc.cx(sys[i], sys[i+1])
```

Listing 1: Controlled Trotter step for the Ising Hamiltonian (one slice).

```python
# Start ancilla in |1>
qc.x(qr_anc)
for m in range(cycles):
    t, dt = t_list[m], t_list[m] / TROTTER
    qc.h(qr_anc)

    # ... Trotter steps inserted here ...

    qc.p(E_target * t, qr_anc)
    qc.h(qr_anc)
    qc.measure(qr_anc, cr_anc)

    # Conditional reset: undo phase and flip back if measured |1>
    qc.p(-E_target * t, qr_anc)
    qc.x(qr_anc)
```

Listing 2: Single-ancilla rodeo filter with mid-circuit measurement and conditional phase reset for hardware execution.

```python
# Controlled forward hopping
qc.crz(-2 * hop, anc, spin)
# Multi-controlled position increment
ctrl_list = [anc] + list(pos[:-1])
qc.mcx(ctrl_list, pos[-1])
for i in range(len(pos) - 2, 0, -1):
    ctrl = [anc] + list(pos[:i])
    qc.mcx(ctrl, pos[i])
qc.cx(anc, pos[0])
```

Listing 3: Increment logic for the Dirac kinetic hopping term.

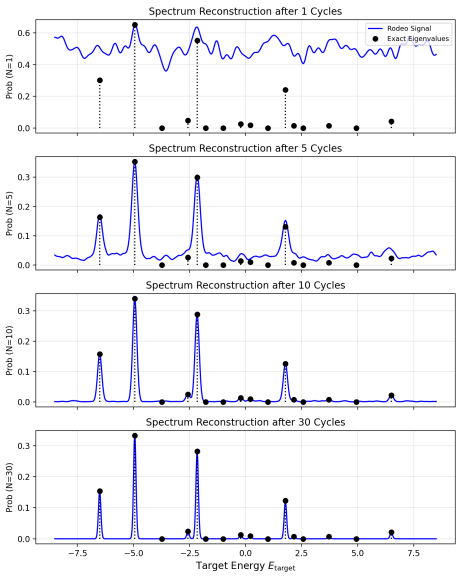

Figure 2: Reconstruction of the spectrum dynamically vs theoretical Hamiltonian eigenvalues (dashed stems) for the initial state $|0000\rangle$. At low N, the signal is highly oscillatory; as $N \to 30$, distinct exponential-decay peaks align exactly with the true eigenvalues.

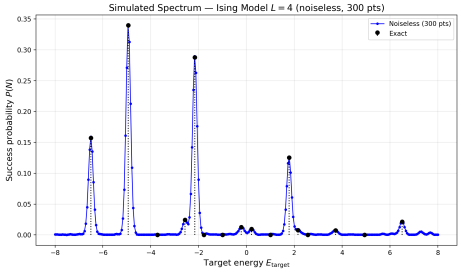

Figure 3: Noiseless rodeo spectrum for the Ising model (L=4, J=1, h=1.5). Ten rodeo cycles, 300 scan points, 3000 shots equivalent.

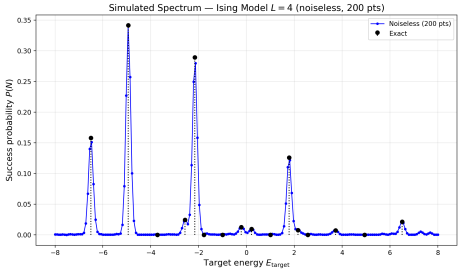

Figure 4: Same parameters as Fig. 3 but with 200 scan points, showing consistent peak positions.

yet the dominant eigenvalues remain identifiable as local maxima. The realistic IBM Manila profile introduces additional readout and coherence errors, further flattening the spectrum but preserving the qualitative peak structure for the lowest-lying states.

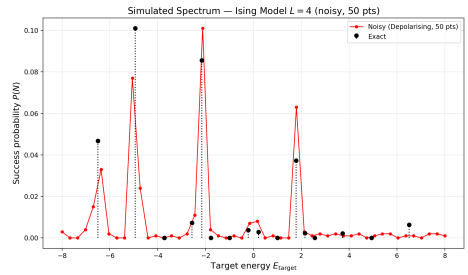

Figure 5: Noisy rodeo spectrum with custom depolarising errors (*noise_level*=0.15); 10 cycles, 50 points.

4.3 Discrete Dirac Spectrum

The generalised circuit for the lattice Dirac Hamiltonian (Eq. 4) with L=4 sites and mass m=1.0 produces spectral peaks consistent with the expected dispersion relation $E = \pm\sqrt{(\sin k\,a/a)^2 + m^2}$.

Crucially, the algorithm extracted both positive energy *matter* and negative energy *antimatter* branches effortlessly (Fig. 6). In this first-quantized single-particle formalism, the negative-energy *antimatter* states naturally emerge as intrinsic eigenvalues of the matrix. The target eigenstate spectrum is thus drawn purely by scanning E_{target}, cleanly capturing the full spectrum without invoking the second-quantization ladder operators of Quantum Field Theory.

While the deeper circuits required here make the method more sensitive to noise, the noiseless simulation cleanly resolves the particle and anti-particle branches (Fig. 6) exactly at the theoretical analytic positions, adjusting dynamically to parametric sweeps of the fermion mass m.

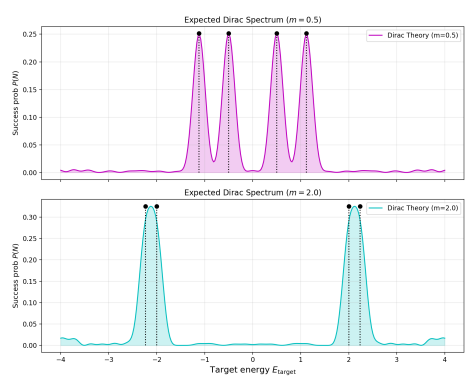

Figure 6: Expected rodeo spectrum for the 1D discrete Dirac Hamiltonian under two distinct mass parameters (m = 0.5 top, m = 2.0 bottom), showing accurate antimatter and matter eigenvalue extractions.

4.4 Alternative Test: Heisenberg XXX Chain

To further demonstrate the algorithm's versatility beyond the Ising and Dirac regimes, we applied the filter to the 1D isotropic Heisenberg XXX anti-ferromagnet ($H = J \sum \vec{\sigma}_i \cdot \vec{\sigma}_{i+1}$) for $L = 3$ sites ($J = 1.0$). The theoretical Pauli matrix construction for the open-boundary XXX model is explicitly defined in Listing 4.

```
def get_heisenberg_hamiltonian(L, J=1.0):
    dim = 2**L
    H = np.zeros((dim, dim), dtype=complex)
    X = np.array([[0, 1], [1, 0]], dtype=complex)
    Y = np.array([[0, -1j], [1j, 0]], dtype=complex)
    Z = np.array([[1, 0], [0, -1]], dtype=complex)
    I = np.eye(2, dtype=complex)

    def get_op(op_type, site):
        op = np.eye(1, dtype=complex)
        for j in range(L):
            op = np.kron(op, op_type if j == site else I
                )
        return op

    for i in range(L - 1):
        H += J * np.dot(get_op(X, i), get_op(X, i+1))
        H += J * np.dot(get_op(Y, i), get_op(Y, i+1))
        H += J * np.dot(get_op(Z, i), get_op(Z, i+1))
    return np.real(H)
```

Listing 4: Constructing the dense Hamiltonian matrix for the 1D Heisenberg XXX spin chain.

The algorithm (Fig. 7) effectively resolves the highly degenerate multiplet structure expected from SU(2) symmetry.

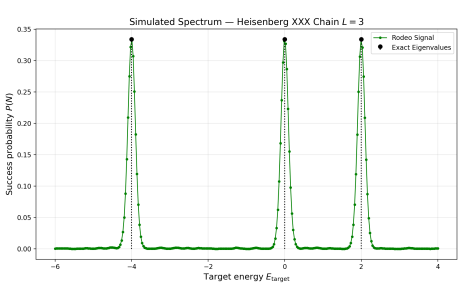

Figure 7: Rodeo spectrum for the $L = 3$ Heisenberg XXX model. Continuous red lines show the simulated filter, exactly intersecting theoretical eigenvalues (stems).

5 Conclusions

We have demonstrated a complete Qiskit-based implementation of the rodeo algorithm for quantum eigenvalue estimation. Key findings:

1. The noiseless rodeo filter resolves the full eigenvalue spectrum of the L=4 transverse-field Ising model with only 10 ancilla cycles and ∼20 000 shots per point.

2. Depolarising noise at realistic gate-error levels (∼1% for two-qubit gates) broadens peaks but preserves the dominant spectral features. IBM hardware-level noise further degrades the signal, motivating error-mitigation strategies.

3. The algorithm generalises naturally to lattice fermion Hamiltonians: we presented the first rodeo-based spectral scan of the one-dimensional discrete Dirac operator.

As an **intermediate presentation summary**, our goals for the future phases of this research involve:

- **Noise Robustness Limits:** We aim to stress-test the noisy algorithm iteratively to find the critical gate-fidelities where the spectral signal is completely drowned in decoherence. Currently, our noisy execution showed signal retention, but the absolute limits without error mitigation remain unknown.

- **2D Dirac Fermions:** We will expand the Hamiltonian from 1D to a 2D square lattice to observe full antimatter propagation and higher-dimensional spectrum dynamics.

References

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